HOCKEY GOALIE STATS

DATA DRIVEN DEVELOPMENT

A goalkeeper's performance is often hard to track and very often does not follow the team's performance, i.e., a goalie can play the best game of his/her life, but if the team doesn't score, the team can still lose the game. This can be demoralizing for the goalie if you are only tracking goals conceded. Data-driven statistic tracking can change the outlook from feelings about the game to objective takeaways that can be turned into training regiments and track improvements. No longer dealing with guesswork but patterns of prevention/interventions of the attack, transforming the goalie into a strategic asset rather than a purely reactive player.

Some important concepts to clarify:

Shot Situation describes the game context in which the shot occurred. Normal Game Play refers to a standard open-play shot, while Penalty Corner includes any first shot or rebound arising during a penalty-corner phase. Penalty Stroke refers to a formal penalty stroke, and 8-Second 1v1 refers to an eight-second, shootout-style attempt.

Shot Type describes the kind of shot taken within that situation. A 1st Shot is the first attempt in an attack, while a Rebound is a subsequent attempt following an initial save or deflection. In-Game 1v1 refers to a normal match-play one-on-one attempt rather than an eight-second shootout. Own Goal refers to a ball scored by a player's teammate into their own goal, while Other covers any shot that does not fit the listed categories.

Saves are recorded when the goalie directly prevents a shot or scoring attempt that would reasonably have resulted in a goal without the goalie's intervention. This can include stopping the ball with the pads, kickers, stick, gloves, body, or any controlled defensive action by the goalie. A Save should only be recorded when the attempt is genuinely on target or dangerous enough that the goalie's action prevents a goal. It should not be recorded when the attacker simply misses the goal, shoots wide, or loses control without the goalie causing the outcome. Saves are used in the Save Rate calculation and also form part of the Defense Rate calculation.

Angle Closed Off should be recorded when the goalie's movement, positioning, or presence removes the attacker's realistic scoring angle and forces a low-quality attempt, a missed shot, or no shot at all. It should not be used when the attacker simply misses an open chance through poor aim. This stat is important because it recognizes that a goalie's job is not only to save shots, but also to prevent dangerous scoring chances from developing. Angle Closed Off is included in Defense Rate, but not in Save Rate, because it is a successful defensive intervention rather than a direct save. In heat maps, Angle Closed Off is shown on the D Heat Map only, because there is no goal-box landing point to record.

- Angle Closed Off is included in **Defense Rate**.
- Angle Closed Off is not included in **Save Rate**.
- Angle Closed Off appears on the D heat map only, because there is no goal-box landing point.

Outnumbered Attempts are recorded when the goalie faces an attacking situation where the number of attackers is greater than the defensive cover available in the immediate scoring area. Examples include 2 vs 1, 3 vs 1, or 4+ vs 1 situations. Tracking this is important because goals conceded while outnumbered may not always reflect a technical failure by the goalie, but may point to a defensive structure problem in front of the goalie. A useful measurement is **Rate of Attempts at Goal Outnumbered % = Outnumbered Attempts at Goal ÷ Total Attempts at Goal × 100**. For this calculation, attempts at goal should include Saves and Goals Conceded, but not Angle Closed Off, because Angle Closed Off is a prevented scoring chance rather than a direct attempt on goal.

Consistent data entry is important if the stats are going to produce useful coaching insights. Every shot record should have one clear outcome: Save, Goal, or Angle Closed Off. Every shot should also have both a Shot Situation and a Shot Type, with Normal Game Play used as the default Shot Situation when no special situation applies. Rebound Result should only be used where relevant, especially after saves, and should identify whether the rebound went to a Safe or Dangerous area. Penalty Stroke attempts should always be recorded from the penalty spot on the D Heat Map. Angle Closed Off does not require a goal-box placement, because the ball did not create a normal goal-box save or goal location. Shootout data should be kept separate from normal match-play data so that reports remain accurate.

Data collected can be separated into two main categories: performance measuring and data to assist with training programs.

Performance Measuring Data:
Three stats that are measured in this category are save rate, defense rate, and rebound distribution rate.

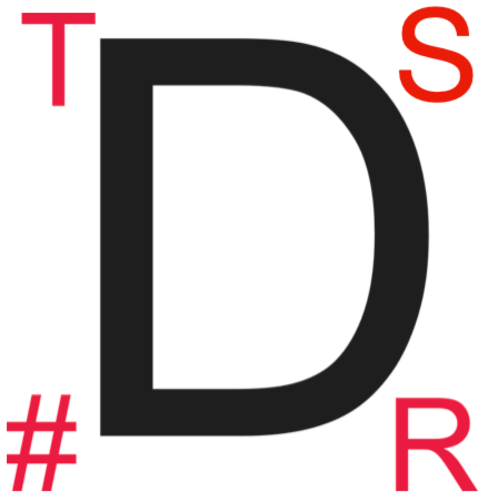
Save Rate:
Simply put, this is the percentage of shots on target faced (total scoring attempts that would have resulted in a goal without intervention) that were saved.
Save Rate % = (Saves ÷ Shots on Target) × 100
If you want to take it further, also differentiate between the first shot saved and saves/goals of the rebound to see if the goalie is recovering fast enough from the initial save.

Defense Rate
It can be said that the best save you can make is the one you don't have to! That is, often a goalie can close off the attacker's possible angle of attack, and the shot is taken missing the goals because the goalie had the goals covered. This save should be accredited to the goalie; this does not include shots where a goal was possible, but the attacker had bad aim.
Defense Rate % = **((Saves + Angles Closed Off) ÷ Total Shot Records) × 100**, where Total Shot Records = Saves + Goals Conceded + Angles Closed Off.

Rebound Distribution Rate
This arises from the thought that the goalie is not purely a defensive player but in fact can start the next offensive movement. For this, you record on every save whether the rebound went to a safe zone or a dangerous zone. A safe zone being either directly to an unmarked teammate or to an open area either side of the goals (not directly in the strike zone). Safe zone deflections + dangerous zone deflections should always = the total number of saves.
Rebound Defense Rate % = (Safe Zone deflections ÷ Total Number of Saves) × 100

Other interesting statistics to record if possible include:
Goals & saves outnumbered, i.e., multiple attacks against the goalie
Goals & saves from penalty corners
Goals & saves from penalty strokes
Goals & saves from 8-second 1v1
Goals & saves from goalie challenging attacker 1v1 in game
Own goals scored by teammates

Suggested Method of Data Collection
If recording stats manually during a game, a shorthand for data collection is suggested to keep up with the fast-paced game. Recording save data first, then graphical data used for training second. This shorthand is a data entry for each attempt at goals. Then later, after the match, use the data collected to collate stats for the game.



D: Main Data Point
- S = Save
- G = Goal
- A = Angle Closed Off

S: Shot Situation
- “” = Normal Game Play
- PC = Penalty Corner
- PS = Penalty Stroke
- V = 8 Second 1v1

T: Type of Shot
- “” = 1st Shot at Goal
- R = off Rebound
- C = Challenged Attacker 1v1
- O = Own Goal scored by teammate

#: Number of Attackers VS Goalie in D

R: Rebound Zone
- S = Safe
- D = Dangerous

Performance Benchmarking and Analysis
Raw data only results in coaching insights when interpreted through a framework of benchmarks. Suggested performance benchmarks.

Level	Expected Save %
U13	65–75%
U16	70–80%
1st Team	75–85%
Provincial	80%

If possible, don’t view on Match as a standalone; rather, look at a 5-match trend.

The Interpretative Framework:
High Saves + High Goals: This indicates a systemic failure in the team’s defensive structure, allowing an excessive volume of high-quality entries.

Low Save %: Typically points to individual technical lapses, such as late reactions, poor positioning, or a failure to be “set” before the shot.

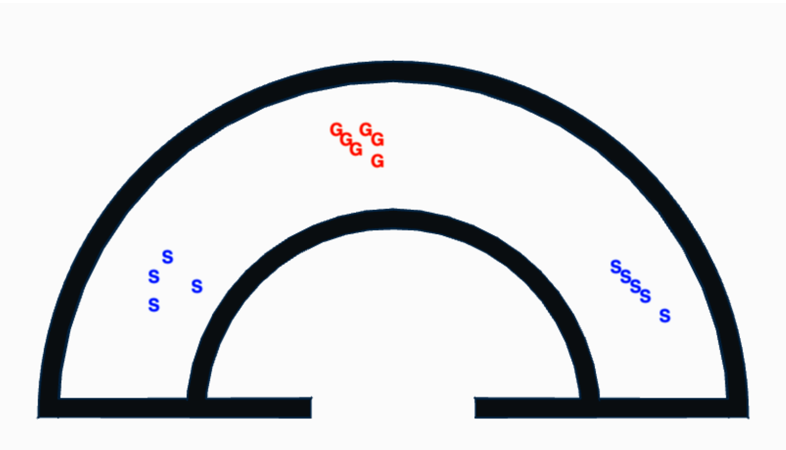
Frequent Rebound Goals: Indicates a need for a concentrated focus on kicking width and clearing to the sidelines.

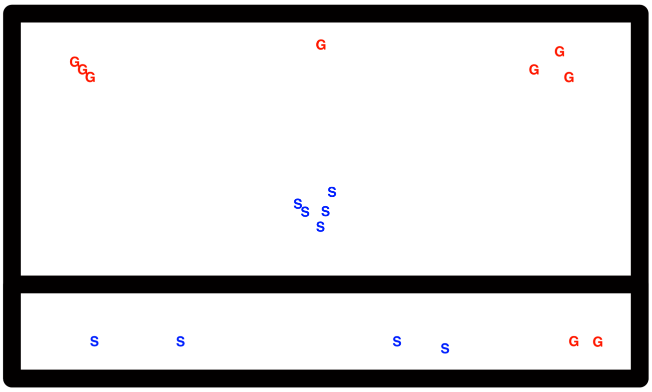
Qualitative Command and Organization
Subjective coaching ratings provide context to the numbers. Evaluating the goalie as the “defensive organizer” whose vocal presence reduces the overall shots faced.

The 1–5 Performance Rating Scale:
Defensive Organization: Effectiveness in directing markers and managing the structural integrity of the back line.
Penalty Corner Communication: Clarity and timing of instructions during the set-piece setup and execution.
Overall Presence: The ability to remain loud, confident, and physically imposing to the opposition.
Consultant Intervention: A score of “1 / 2 requires an immediate tactical intervention. The goalkeeper should call out marking assignments at least 5 seconds before circle entry to ensure the defensive line remains efficient and proactive.

Data to assist with training programs
This requires a different type of data to be recorded. This is the How, not the What.
Data points to record are **Where was the shot taken from** Record from which area in the D was the shot taken from and whether it was a goal or save. This will show a pattern of which areas of the D the goalie struggles to save shots from, and coaches should train shots from this area. This can be recorded visually by marking a diagram of the D and creating a heat map.

While one match’s data can show a specific team’s shooting style/abilities, after a few matches, a trend may be visible, and a training program can be developed to work on areas that goals are often scored from. **Where the Shot lands** Similarly, by marking where in the goal box the goal is scored, a training program can be developed to work on weaker areas.





Shootout / 8 Second 1v1 Summary Stats
Shootout attempts and 8 Second 1v1 situations should be analysed separately from normal match-play shots because the tactical situation is different. In a shootout, the goalie faces a structured 1v1 from a set starting point, whereas normal match-play shots develop from open play, penalty corners, rebounds, or defensive breakdowns. For this reason, shootout data should be shown as a summary rather than mixed directly into normal shot totals. Useful shootout stats include Shootout Rounds, Shootout Saves, Shootout Goals Conceded, Shootout Save Rate, Shootout Defence Rate, and Most Common Save Method. This allows the coach to evaluate the goalie’s 1v1 performance without distorting normal match statistics.